

Response to Reviewers

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Title: Point Mass Planning and NMPC Tracking for Time-Optimal Agile Quadrotor Gate Navigation

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We thank the Associate Editor and both reviewers. Both found the work technically sound and a contribution to the field, and we have addressed every comment. Changes are marked in yellow in the highlighted PDF; section/equation labels refer to the revised manuscript. No new references were added.

Reviewer 1

The reviewer notes we identified and addressed two significant research gaps, and answered *Yes* to contribution, technical soundness, and reference adequacy. We address the two suggestions below.

Comment 1.1 — more detailed explanation of the concepts (Q2)

Concern. The planner/controller description, while sufficient, could be more detailed.

Response. We added insight at the one point previously stated without motivation—the point-mass cost structure—as it links the planner to the quality of the reference passed to the tracker.

Action. A sentence after Eq. (PMM cost, Sec. III-C) now explains the three cost terms, in particular that the jerk regulariser (w_j) keeps the downstream flatness derivatives (attitude, body rate) well conditioned, since they are obtained by differentiating the planned acceleration.

Comment 1.2 — criterion for the NMPC parameters N , f_c (Q3)

Concern. No criterion is given for selecting the NMPC parameters.

Response. The values follow standard practice and the real-time budget, not a separate tuning study, which we now state. $f_c = 100$ Hz is the conventional rate for body-rate agile NMPC; $N = 50$ ($T_h = 0.5$ s) is the largest horizon keeping the P99 solve time within the 10 ms deadline on the test platform, and the half-second look-ahead spans roughly one inter-gate transit ($T_f^*/N_g \approx 0.7$ s).

Action. Added an explanatory passage in Sec. V-A after the parameter list, giving the criterion for f_c and N/T_h and linking it to the measured real-time budget (Sec. V-G).

Remaining items

Questions 1, 4, and 5 were affirmative and required no action. No references were suggested for addition or removal.

Reviewer 2

The reviewer found the paper “very well written, well-founded and referenced” and recommended publication after minor adjustments.

Comment 2.1 — discuss the contribution against similar cited works

Concern. Include a brief discussion positioning the proposal against similar cited works, highlighting its advantages.

Response. We agree; the paper did not consolidate this comparison in one place. We added it using only references already in the manuscript.

Action. A paragraph at the end of Sec. V-H (Discussion) now positions the work: it retains the fast point-mass planning of Foehn et al. but feeds the *complete* flat output to the NMPC; it keeps planner and tracker modular—unlike the fused MPCC of Romero et al. and Krinner et al., whose stages cannot be replaced independently—while recovering the tracking completeness those methods achieve implicitly; and, unlike Sun et al., it *ablates* the reference structure to isolate which flat components drive gate-crossing performance, at no extra solver cost.

Remaining items

All four assessment questions were affirmative; no references were suggested for addition or removal.

Additional corrections by the authors

We audited every quantitative claim against the raw result files. Tables III and IV, all reported percentages, peak speeds, and peak accelerations were confirmed correct. Two items were corrected:

AC1 — Blow-up exclusion.

The text gave an approximate per-mode divergence rate. We replaced it with the exact paired-exclusion outcome: trials are dropped if either controller diverges ($\text{RMSE} > 5 \text{ m}$), leaving 23/25 valid trials on the figure-8 and 18/25 on the vertical loop (Sec. V-D).

AC2 — Figure 2.

The original figure had been rendered from an earlier planner export and the model-in-the-loop set, so its numbers did not match the software-in-the-loop tables. It is regenerated from exactly the reported data (Table III reference; software-in-the-loop trajectories).

We believe the revision fully addresses the reviewers' comments and thank the reviewers and Associate Editor.